

Civilizational Intent Conservation: A Proposed Framework for Preserving Collective Intent in Civilization-Scale AI Systems

Shaurya Sanyal (Team Auralis)

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Abstract

As AI systems scale toward civilization-level coordination, a critical question emerges: can such systems maintain fidelity to their originally articulated objectives, or will optimization pressure cause unintentional drift? This paper proposes Civilizational Intent Conservation (CIC) - a framework for detecting, auditing, and governing changes to collective societal intent within AI-mediated systems. CIC distinguishes between unintentional intent drift and legitimate, explicitly authorized intent evolution. We define the architecture, threat model, proposed benchmark (CIC-001), and formal metrics. No experimental results are reported; this paper establishes the theoretical foundation.

1. Introduction

- The alignment problem at civilization scale
- Current alignment research focuses on individual model behavior
- CIC extends alignment to institutional/societal intent continuity
- Key question: 'Can a civilization-scale intelligent system become increasingly capable without silently losing the intent it was created to serve?'
- CIC is a PROPOSED research framework, not a solved problem

2. Definition

- Civilizational Intent Conservation = preserving continuity and provenance of collective societal intent while allowing explicitly authorized evolution
- Unintentional intent drift vs legitimate authorized intent evolution
- Example: original intent 'human flourishing + ecological sustainability' gradually optimized to 'economic output' without authorization
- CIC is NOT value policing - it does not decide what values should be

3. CIC Architecture

- Intent Repository: versioned objectives, principles, constraints, provenance, authorization history
- Intent Graph: directed graph of values, objectives, constraints, priorities, institutions
- Value Auditor: compares intended objectives with observed system behavior
- Drift Detector: statistical change-point detection on intent metrics (objective drift, priority drift, proxy drift, semantic drift, institutional drift)
- Intent Evolution Gate: determines if change was accidental/emergent/manipulated/proposed/authorized
- Intent Provenance Recorder: logs WHO, WHAT, WHEN, WHY, AUTHORIZATION, EVIDENCE, PREVIOUS VERSION, NEW VERSION, SIMULATED CONSEQUENCES

4. CIC Safety Principle

- CIC operates: OBSERVE -> COMPARE -> DETECT -> EXPLAIN -> SIMULATE -> ESCALATE -> RECORD -> REQUEST AUTHORIZATION
- CIC must NOT: OBSERVE -> DECIDE WHAT HUMANS SHOULD VALUE -> FORCE COMPLIANCE
- Major intent changes require explicit governance/human/civic authorization
- Integrates with OPA, HITL, CHRONOS, geofence, kill switch

5. Integration with ORION

- Flow: SENSE -> OMNIS -> NEXUS -> FORGE -> MIRROR -> ASCEND -> GOVERNANCE -> CIC INTENT AUDIT -> AUTHORIZED ACTION -> OBSERVE -> LEARN -> UPDATE OMNIS -> CIC REASSESSMENT -> REPLAN
- CIC runs both pre-decision and post-decision

6. Related Concepts

- AICC (AI-Mediated Cultural Continuity): cultural memory preservation - supporting research direction, ties to PHOENIX and OMNIS
- EGE (Evolutionary Governance Engineering): simulation-based governance exploration - research/simulation capability, NOT autonomous political authority
- Relationship: AICC = cultural memory, PHOENIX = institutional continuity, OMNIS = world model, CIC = intent continuity

7. Proposed Benchmark (CIC-001)

- Simulated civilization with population, institutions, AI agents, economy, culture, laws, resources, environment
- Seed INTENT v1.0
- Expose to: optimization pressure, generational turnover, institutional changes, AI replacement, economic shocks, resource scarcity, technological breakthroughs, misinformation, cultural evolution, conflicting objectives
- Measure 10 dimensions: intent preservation, drift detection, false positive rate, false negative rate, legitimate evolution recognition, provenance integrity, recovery effectiveness, time-to-detection, intervention cost, governance violations
- STATUS: SPECIFICATION ONLY - no results yet

8. Proposed Metrics

- Intent Continuity (IC)
- Intent Drift Magnitude (IDM)
- Intent Detection Recall (IDR)
- False Drift Rate (FDR)
- Authorized Evolution Integrity (AEI)
- Intent Provenance Integrity (IPI)
- Intent Recovery Effectiveness (IRE)
- STATUS: DEFINITIONS ONLY - no numerical results

9. Threat Model

- Optimization drift (proxy alignment failure)
- Gradual institutional capture
- Adversarial intent manipulation
- Model interpretation drift
- Generational value evolution without explicit authorization
- Semantic drift in objective definitions

10. Limitations

- Entirely theoretical - no experiments conducted
- Benchmark is proposed, not executed
- Formal metrics lack empirical validation
- Integration with ORION is architectural, not implemented
- CIC itself could become a single point of failure if misconfigured

11. Conclusion

- CIC addresses a gap between individual-model alignment and civilization-scale intent preservation
- The framework is proposed as a research direction within the ORION architecture
- Next step: implement minimal CIC prototype and execute CIC-001 benchmark

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